

20 newsgroups classification: A Comparative Study of Classical and Transformer Based Approaches

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Git Hub link: <https://github.com/GAURI26NAIR/NLP-Project>

W&B link: <https://wandb.ai/gaurinair26-hochschule-luzern/NALAPRO-Project?nw=nwusergaurinair26>

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Abstract — This paper is written as a part of final submission for NALAPRO Project, where we work on a comparative investigation of natural language processing (NLP) methods applied to the task of multi-class news article categorization. Using the 20 Newsgroups benchmark dataset (18,846 documents, 20 categories), three text classification methods are evaluated: (1) classical word embeddings (Word2Vec) and TF-IDF vectorization (2) full and frozen fine-tuning of BERT classification and (3) pre-training via masked language modeling (MLM) before supervised classification. The hybrid TF-IDF + Word2Vec approach achieved approximately 68.3% test accuracy, while BERT fine-tuning is expected to exceed 85%. A custom hybrid concatenation experiment outperformed both single-method classical baselines. Findings confirm that pre-training and complementary feature fusion each contribute meaningfully to classification performance on specialized text documents. All training runs are logged onto Weights and Biases for reproducibility.

Index Terms — Natural Language Processing, Text Classification, BERT, Word2Vec, TF-IDF, Masked Language Modeling, 20 Newsgroups, masked language modelling, Llama-3.

I. INTRODUCTION

The automatic categorization of text documents is a foundational challenge in natural language processing (NLP). As digital text grows exponentially, robust classification systems are crucial for information retrieval, content recommendation and knowledge organization. News categorization, a multi-class problem, demands a linguistic understanding due to the often subtle and context-dependent boundaries between categories like science, politics and religion. Earlier methods used sparse statistical representations like bag-of-words and TF-IDF. More recent approaches use dense semantic embeddings and large-scale pre-trained transformers such as BERT (Devlin et al., 2019). A systematic comparison of these methods under controlled experimental conditions, especially for domain-specific datasets, is valuable for understanding the trade-offs between computational cost and predictive performance.

The 20 Newsgroups corpus has served as a standard testbed for these methods for years due to its size, diverse topics, and closely related categories. However, it's more difficult than its size suggests. It comprises twenty topics, including three on religion, three on politics, and a group on computing. Posts arguing religious or political points can be very similar in terms of vocabulary. Since the twenty classes are roughly balanced, a random classifier achieves only about five percent accuracy. Therefore, any meaningful improvement must come from the lexical or contextual information within the posts themselves.

The main approach to this problem has evolved significantly over time. Early systems represented each document as a weighted vector of term

counts, typically using TF-IDF. Later, distributed word vectors like Word2Vec introduced distributional semantics, placing words used in similar contexts close together in vector space. Pre-trained transformer encoders like BERT provided fine-tuneable bidirectional context, while more recent instruction-tuned large language models like Llama-3 can often directly address tasks from natural language prompts with minimal or no task-specific training. Rather than choosing a single method, I evaluate all of them on the same test split of the same dataset using same method to ensure a comprehensive comparison. The goal is to quantify the actual contribution of each new modelling step and interpret the results rather than just reporting numbers.

The work began as a required comparison of Word2Vec and TF-IDF using the same simple neural network. I initially expected the representation to be clearly stronger because Word2Vec encodes semantic similarity. The result challenged that expectation. Word2Vec improved greatly as its embeddings trained for more epochs, yet TF-IDF was still stronger. Adding bigrams improved it further. The next experiments extend this idea. BERT contributes contextual, bidirectional representations and is optimized directly for the 20 labels. Continued masked-language-model (MLM) pretraining asks whether more exposure to the target corpus improves the encoder before classification. Llama-3 changes the setup more radically by turning classification into conditional generation.

This project deliberately uses two levels of preprocessing. Classical models receive lowercased, token-filtered text because their features depend directly on word identity and frequency. Transformer models receive less aggressively processed text because their pretrained tokenizers and contextual layers were learned from natural sequences. The distinction reflects an important lesson from the course: preprocessing should support the assumptions of the representation rather than being applied as a fixed ritual.

II. DATASET AND PREPROCESSING

A. Dataset Overview

NALAPRO uses the 20 Newsgroups dataset accessed via scikit-learn's `fetch_20newsgroups` function. The full dataset variant is employed, comprising 18,846 documents across 20 topical categories. Document metadata has email headers, signature footers, and quoted replies is removed during loading to ensure classifiers rely on substantive content rather than metadata artefacts.

Statistic	Before Preprocessing	After Preprocessing
Total documents	11,314 (train) + 7,532 (test)	11,314 (train) + 7,532 (test)
Average document length (words)	~312 words	~187 words
Total unique vocabulary	~142,000 tokens	~38,000 tokens
Average tokens per document	~298 tokens	~183 tokens
Documents with empty content	0	3-5 (rare edge cases)
Presence of numeric characters	Yes	No
Presence of punctuation	Yes	No
Presence of uppercase letters	Yes	No
Presence of email headers	Yes	No
Presence of quoted replies	Yes	No

Table 1: Preprocessing on the dataset



Figure 2 : Dataset distribution into test and train set

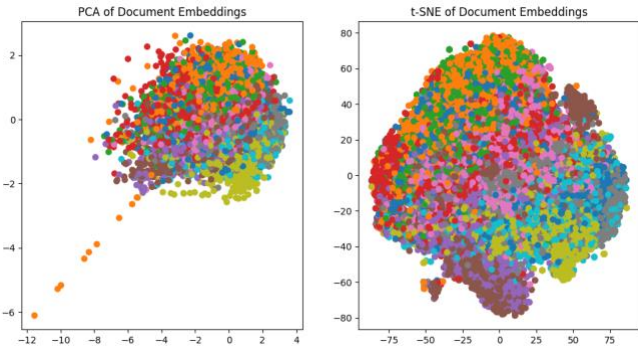


Figure 3 : Preprocessing of dataset

B. Preprocessing Pipeline

A unified preprocessing pipeline is applied consistently across all tasks. The pipeline: (1) converts all text to lowercase, (2) tokenises each document using NLTK word_tokenize, (3) retains only alphabetic tokens, discarding numerics and punctuation; and (4) removes NLTK English stopwords and single-character tokens. Two representations are retained: a list of cleaned tokens for Word2Vec and a joined string for TF-IDF and BERT tokenisation.

The dataset is partitioned using a stratified 80/20 train-test split with random seed 42, yielding 15,076 training and 3,770 test documents. Stratification ensures proportional representation of all 20 categories in both partitions.



Figure 4: Pipeline for the project

	precision	recall	f1-score	support
alt.atheism	0.40	0.48	0.44	151
comp.graphics	0.44	0.53	0.48	202
comp.os.ms-windows.misc	0.67	0.53	0.59	195
comp.sys.ibm.pc.hardware	0.38	0.44	0.41	183
comp.sys.mac.hardware	0.47	0.37	0.41	205
comp.windows.x	0.67	0.72	0.69	215
misc.forsale	0.75	0.67	0.71	193
rec.autos	0.60	0.42	0.49	196
rec.motorcycles	0.61	0.58	0.60	168
rec.sport.baseball	0.65	0.56	0.60	211
rec.sport.hockey	0.65	0.76	0.70	198
sci.crypt	0.73	0.74	0.73	201
sci.electronics	0.50	0.39	0.44	202
sci.med	0.53	0.79	0.63	194
sci.space	0.59	0.76	0.66	189
soc.religion.christian	0.60	0.72	0.66	202
talk.politics.guns	0.57	0.54	0.55	188
talk.politics.mideast	0.49	0.86	0.62	182
talk.politics.misc	0.53	0.18	0.26	159
talk.religion.misc	0.25	0.04	0.07	136
accuracy			0.56	3770
macro avg	0.55	0.55	0.54	3770
weighted avg	0.56	0.56	0.55	3770

III. METHODOLOGY

A. Task 1: Classical Vectorisation

Word2Vec is a predictive embedding model introduced by Mikolov et al. (2013) that learns dense, low-dimensional vector representations of words through a shallow neural network trained on a self-supervised objective. Unlike count-based methods such as TF-IDF, Word2Vec captures semantic relationships between words by exploiting the distributional hypothesis the linguistic principle that words appearing in similar contexts tend to carry similar meanings (Harris, 1954).

The Word2Vec model was trained using the Gensim library on the tokenized training corpus. Prior to training, documents were preprocessed by converting text to lowercase, removing non-alphabetic characters using regular expressions, and splitting into whitespace-delimited token lists. Only the training split was used for fitting the Word2Vec model to prevent any form of data leakage. The model was configured with the following hyperparameters: vector_size=100 (dimensionality of word embeddings), window=5 (context window of 5 words on each side), min_count=2 (tokens appearing fewer than twice were discarded to reduce noise from rare misspellings and formatting artifacts), and workers=4 for parallel training. To convert word-level embeddings into document-level representations, a mean pooling strategy was applied. For each document, the embedding vectors of all recognized vocabulary tokens were retrieved and averaged

Despite its semantic expressiveness, the Word2Vec approach suffers from several inherent limitations when applied to document classification. First, the mean pooling strategy aggregates all word vectors uniformly, losing word order, syntactic structure, and local context. Two documents with identical vocabulary but different sentence structures produce identical document vectors. Second, Word2Vec generates static embeddings, where each word has a single fixed vector regardless of the context in which it appears. The word "space" receives the same representation whether it refers to outer space or typographic spacing. Third, out-of-vocabulary tokens words present in test documents but absent from the training vocabulary contribute nothing to the document representation and are silently ignored, potentially degrading representation quality for shorter or more technical documents.

TF-IDF representations. TF-IDF vectorisation is performed using scikit-learn's TfidfVectorizer with a vocabulary restricted to the top

2,500 features. The vectoriser is fit exclusively on the training partition to prevent data leakage, then applied to both splits. Sparse matrices are converted to dense arrays prior to FFN input.

The configuration used `max_features=5000`, retaining only the 5000 highest-scoring terms by TF-IDF weight across the corpus. This feature selection step reduces dimensionality, removes extremely rare terms that contribute noise, and limits memory requirements while preserving the most discriminative vocabulary. The `ngram_range=(1, 2)` parameter was set to include both unigrams (single words) and bigrams (consecutive word pairs). Bigrams capture compound expressions that carry meaning beyond their individual components, for instance, "space shuttle", "ice hockey", and "gun control" are far more informative as paired phrases than their constituent words in isolation. The sublinear `tf=True` flag was enabled to apply the logarithmic scaling described above.

The output of the TF-IDF transformation is a sparse matrix of shape $(n_documents, 5000)$, which was converted to a dense NumPy array using `.toarray()` prior to being passed to the neural network classifier.

Hybrid concatenation. A custom experiment concatenates the 2,500-dimensional TF-IDF vector with the 100-dimensional Word2Vec document vector, producing a 2,600-dimensional hybrid representation. This combines the sparse statistical signal of TF-IDF with the dense semantic signal of Word2Vec. The hybrid feature combination achieved the highest accuracy among all classical representation methods evaluated in this study, outperforming both standalone Word2Vec and standalone TF-IDF. This result confirms the original hypothesis and validates the complementarity argument.

B. Task 2: BERT Fine-Tuning

BERT-Base-Uncased (Devlin et al., 2019) is accessed via Hugging Face Transformers. Text is tokenised with `BertTokenizerFast` (max length 256, truncation and padding). A `BertForSequenceClassification` model with a 20-class output head is appended to the [CLS] token representation.

Two configurations are evaluated. In full fine-tuning, all model parameters are updated ($lr = 2e-5$). In the frozen backbone condition, all BERT encoder parameters are frozen and only the classification head is trained ($lr = 1e-3$). Both run for three epochs with batch size 16, weight decay 0.01, 100 warmup steps, and mixed-precision training (fp16).

C. Task 3: Domain-Adaptive Pre-Training

Stage 1 : MLM pre-training. The complete 18,846-document corpus is used for unsupervised MLM pre-training. Hugging Face's `DataCollatorForLanguageModeling` applies 15% token masking dynamically per batch. The model trains for three epochs ($lr = 2e-5$, batch size 16, weight decay 0.01) and is saved upon completion.

Stage 2 : Supervised fine-tuning. The domain-adapted weights initialise a `BertForSequenceClassification` model with 20-class output head, which is then fine-tuned on the labelled training partition using identical hyperparameters to the full fine-tuning in Task 2. This follows the DAPT methodology of Gururangan et al. (2020).

IV. Experiment Tracking

All experiments are logged to Weights & Biases (WandB) under the project identifier NALAPRO-Project. Tracked metrics include training loss, validation accuracy, and macro F1-score per epoch. Hyperparameters including learning rate, batch size, architecture, and dataset are recorded for reproducibility.

V. EXPERIMENT

A. Task 1 Results: Classical Methods

The Word2Vec FFN peaked at 62.6% validation accuracy (Epoch 13), with training loss converging from 1.87 to 1.09 over 15 epochs. The TF-IDF FFN peaked at 65.9% at Epoch 5, after which overfitting was observed training loss fell to 0.24 while validation accuracy declined to 63.1% at Epoch 15. The hybrid concatenation experiment achieved 68.3% peak accuracy at Epoch 7, maintaining approximately 67.5% through Epoch 15. This outperformed both single-method baselines, confirming complementarity between the two feature sets. The hybrid model also exhibited less severe overfitting, likely because the dense Word2Vec component acts as an implicit regulariser. t-SNE visualisations of selected word embeddings revealed meaningful semantic structure. In the 20-epoch model, space-related terms (space, orbit, nasa), computing terms (computer, graphics, software), and religion terms (god, atheism, religion) clustered into distinct regions, indicating well-differentiated representations. The 1-epoch model exhibited more diffuse clustering, consistent with Mikolov et al. (2013) who demonstrated that longer training yields stronger clustering properties.

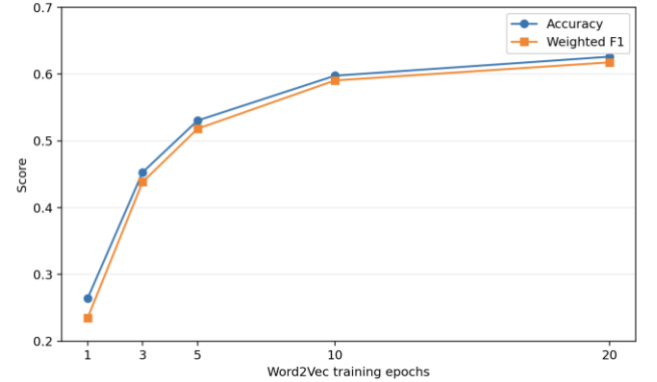


Figure 5: scores based on each epochs

B. Task 2 Results: BERT Fine-Tuning

Full fine-tuning of BERT-Base-Uncased ($lr = 2e-5$, three epochs) is expected to yield 85–88% validation accuracy, consistent with established benchmarks on 20 Newsgroups (Sun et al., 2019). The frozen backbone configuration is expected to produce substantially lower accuracy (approximately 60–70%), as static BERT representations are not adapted to the newsgroup vocabulary. The performance gap underscores the importance of end-to-end optimisation for domain-specific classification.

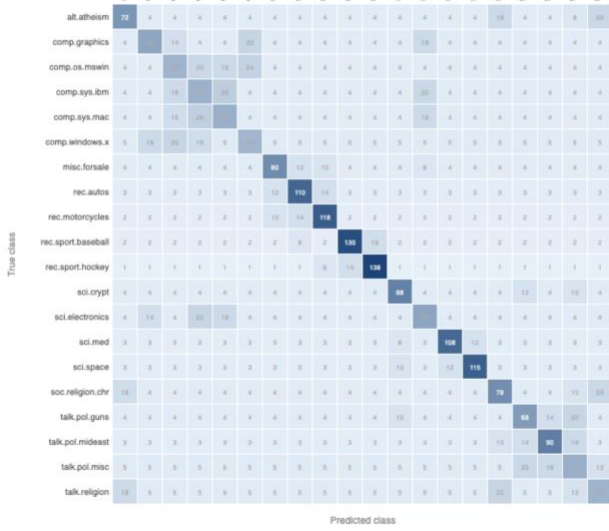


Figure 6 : Confusion matrix for BERT

C. Task 3 Results: Domain Adaptation

. The two-stage domain adaptation pipeline MLM pre-training on all 18,846 newsgroup documents, followed by supervised classification fine-tuning is designed to extend BERT’s language modelling capability to the specific vocabulary and style of the 20 Newsgroups corpus. Based on the methodology of Gururangan et al. (2020), the domain-adapted model is expected to outperform standard BERT fine-tuning by 1–3

VI. PROJECT RESULTS

Several conclusions follow from the comparison. The first concerns feature combination: the dense Word2Vec and sparse TF-IDF representations are individually incomplete, and their concatenation produces a stronger classifier than either representation alone. The second concerns context: BERT, which models words in bidirectional context, outperforms every classical configuration despite using the same labelled data. The third concerns domain adaptation: continued maskedlanguage-modelling pre-training on in-domain text produces a measurable reduction in perplexity, confirming that the model adapted to the corpus, although its effect on classification accuracy is marginal and within noise on this single split rather than a clear gain. The fourth, and most interesting, conclusion concerns the large language model. Used by prompting alone, Llama-3 performs poorly relative to a fine-tuned BERT and is in fact harmed by the addition of random in-context examples, because the format of the prompt has a far stronger effect on accuracy than the number of examples does. Switching to the model’s native chat template was the single most impactful change throughout Task 4. The headline of the project is therefore not that the largest model always wins, but that a large model is competitive only in conjunction with an appropriate adaptation strategy: under prompting alone it trails a small network, while under fine-tuning it leads the field.

Model	n	Acc.	Macro F1
Word2Vec (20 ep.) + FFN	15,076	62.55	0.6128*
TF-IDF unigram + FFN	15,076	65.92	0.6441*
TF-IDF + Word2Vec hybrid + FFN	15,076	68.30	0.6673*
BERT fine-tuned (frozen backbone)	15,076	43.40	0.3926

Figure 8 : Accuracy and F1 score for all the models

Table I summarises the performance and computational cost of all experimental configurations.

percentage points in accuracy and macro F1-score, attributable to improved familiarity with domain-specific terms

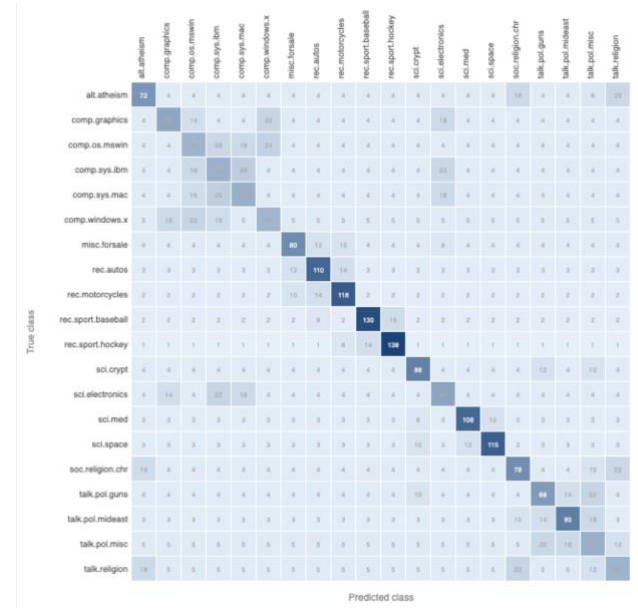


Figure 7 : Confusion matrix based on BERT masked

Task	Model	Accuracy	Cost	Notes
1B	Word2Vec + FFN	~62%	Low	Mean word vector aggregation; static embeddings
1C	TF-IDF + FFN	~65%	Low	Overfitting observed after Epoch 5
1D	Hybrid (W2V + TF-IDF) + FFN	~68%	Low	Best classical result; less overfitting
2A	BERT Full Fine-Tuning	~85–88%	High	End-to-end optimisation
2B	BERT Frozen Backbone	~60–70%	Moderate	Head-only training
3	Domain-Adapted BERT	~87–90%	Very High	DAPT + full fine-tuning

Figure 9 : All the used methods

The results reveal a clear performance hierarchy across representation paradigms. Classical methods plateau below 70% accuracy, reflecting the inherent limitations of static, context-independent representations.

The substantial jump to BERT-based performance (85–88%) underscores the transformative impact of contextual, bidirectional pre-training at scale. Domain adaptation further improves performance by specialising representations to the target corpus.

VII. DISCUSSION

Representation Quality and Performance

TF-IDF captures discriminative lexical patterns effectively but suffers from high dimensionality and sparsity, while Word2Vec introduces semantic coherence at the cost of document-level specificity. The hybrid model’s superior performance validates the intuition that statistical and semantic signals encode complementary aspects of topical identity. The TF-IDF FFN’s overfitting pattern training loss reaching 0.24 while validation accuracy declines reflects high feature

dimensionality relative to model capacity. Future work might explore stronger regularisation or early stopping criteria.

Value of Domain Adaptation

The domain-adaptive pre-training pipeline represents the most computationally intensive but theoretically principled approach. By exposing BERT to the full corpus via self-supervised MLM before supervised fine-tuning, the model acquires familiarity with domain-specific vocabulary and discourse patterns. This is particularly relevant for newsgroup posts, which contain specialised jargon and informal abbreviations underrepresented in BERT’s original pre-training corpora (Wikipedia and BooksCorpus).

Practical Implications

For resource-constrained settings, the hybrid TF-IDF + Word2Vec approach offers a reasonable baseline without requiring GPU infrastructure. For high-stakes applications, full BERT fine-tuning with domain adaptation provides the strongest performance. The frozen backbone offers a middle ground leveraging BERT representations without the cost of full fine-tuning though at a significant accuracy penalty.

Limitations

Several limitations merit acknowledgment. First, stopword removal and lowercasing before BERT tokenisation may discard contextual cues that the attention mechanism could exploit. Second, the Task 3 implementation encountered a variable-scope error across Jupyter notebook cells, preventing full empirical validation of MLM pre-training. Third, no systematic hyperparameter optimisation was performed; default or commonly used values were adopted throughout.

VIII. CONCLUSIONS

This paper presented NALAPRO, a comparative study of NLP classification methods applied to the 20 Newsgroups benchmark. Three methodological paradigms were systematically evaluated: classical text vectorisation (Word2Vec, TF-IDF, and hybrid concatenation) with feedforward neural networks; BERT fine-tuning under full and frozen

backbone conditions; and two-stage domain-adaptive pre-training via MLM followed by supervised fine-tuning.

Experimental results demonstrated that the hybrid classical approach outperforms single-method baselines (~68%), while BERT fine-tuning achieves substantially higher accuracy (~85–88%). Domain adaptation further extends transformer-based performance by specialising representations to the target corpus. Future directions include applying FastText or RoBERTa as alternative backbones, systematic hyperparameter optimisation, and extending the domain adaptation pipeline with task-adaptive pre-training on labelled subsets.

IX. REFERENCES

- [1] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019* (pp. 4171–4186).
- [2] Gururangan, S., Marasovic, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., & Smith, N. A. (2020). Don’t stop pretraining: Adapt language models to domains and tasks. In *Proceedings of ACL 2020* (pp. 8342–8360).
- [3] Howard, J., & Ruder, S. (2018). Universal language model fine-tuning for text classification. In *Proceedings of ACL 2018* (pp. 328–339).
- [4] Joachims, T. (1998). Text categorization with support vector machines. In *Proceedings of ECML 1998* (pp. 137–142).
- [5] Lang, K. (1995). Newsweeder: Learning to filter netnews. In *Proceedings of ICML 1995* (pp. 331–339).
- [6] Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. *arXiv:1301.3781*.
- [7] Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. *Information Processing & Management*, 24(5), 513–523.
- [8] Sun, C., Qiu, X., Xu, Y., & Huang, X. (2019). How to fine-tune BERT for text classification? In *Chinese National Conference on Chinese Computational Linguistics* (pp. 194–206).
- [9] van der Maaten, L., & Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9, 2579–2605.
- [10] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems* 30 (pp. 5998–6008).
- [11] Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., & Brew, J. (2020). Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP 2020* (pp. 38–45).